

DR. BABASAHEB AMBEDKAR TECHNOLOGICAL UNIVERSITY, LONERE

Bachelor of Technology (Artificial Intelligence (AI) and Data Science) SEMESTER - 5 Winter 2025 (Regular)

Course :Bachelor of Technology (Artificial Intelligence (AI) and Data Science) Branch : Engineering and Technology

Semester : SEMESTER - 5

Subject Code & Name: BTAIPE504B - SOFT COMPUTING

Time : 3 Hours]

[Total Marks : 60

Instructions to the Students:

1. Each question carries 12 marks.
 2. Question No. 1 will be compulsory and include objective-type questions.
 3. Candidates are required to attempt any four questions from Question No. 2 to Question No. 6
 4. Use of non-programmable scientific calculators is allowed.
 5. Assume suitable data wherever necessary and mention it clearly.
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Q1. Objective type questions. (Compulsory Question) 12

- 1 Which of the following is NOT a type of soft computing? a) Neural Networks b) Genetic Algorithm c) Compiler Design d) Fuzzy Logic
- 2 The function used to introduce non-linearity in ANN is a) Cost function b) Activation function c) Error function d) Loss function
- 3 Backpropagation algorithm is used for a) Feature extraction b) Forward propagation c) Weight adjustment d) Data normalization
- 4 Which of the following is a feedforward neural network? a) Boltzmann Machine b) Hopfield Network c) Recurrent Neural Network d) Multilayer Perceptron
- 5 PyTorch is primarily used for a) Mobile app development b) Web development c) Deep learning and machine learning d) Database management
- 6 A tensor is best described as a) A vector only b) A scalar only c) A function d) A multi-dimensional array
- 7 Keras is mainly used for a) Data visualization b) Neural network modeling c) Image editing d) System programming
- 8 Which technique helps to reduce overfitting? a) Dropout b) Regularization c) Early stopping d) All of the above
- 9 Fuzzy logic deals with a) Degrees of membership b) Binary values c) Exact reasoning d) Crisp sets
- 10 Membership function value lies between a) 1 to 10 b) -1 to 1 c) 0 to 1 d) 0 to ∞
- 11 Genetic Algorithms (GA) are inspired by a) Fuzzy reasoning b) Mathematical optimization c) Biological evolution d) Human brain
- 12 The quality of a solution in GA is measured by a) Loss function b) Error function c) Fitness function d) Cost function

Q2. Solve the following.

- A) Differentiate between Soft Computing and Hard Computing with suitable examples. 6
- B) Describe the architecture of an Artificial Neuron with the role of weights, bias, and activation function 6

- Q3. Solve the following.
- A) What is Gradient Descent? Explain the working of SGD and Mini-Batch GD. 6
 - B) Explain the need for activation functions in neural networks and its types 6
- Q4. Solve Any Two of the following.
- A) Describe Backpropagation algorithm with a neat diagram. 6
 - B) Explain Feedforward Neural Network with its architecture. 6
 - C) Using McCulloch-Pitts neuron model, design a neural network for 2-input OR functions. 6
- Q5. Solve Any Two of the following.
- A) Explain PyTorch, TensorFlow, and Keras in detail with their features and applications. 6
 - B) Write a short note on overfitting and underfitting in neural networks. 6
 - C) How to split data into training and testing sets? 6
- Q6. Solve Any Two of the following.
- A) Explain Classical Set vs. Fuzzy Set with diagrams. 6
 - B) Describe the operators of Genetic Algorithms. 6
 - C) Illustrate Fuzzification and Defuzzification methods with examples. 6

***** End *****